LESSON 3

Introduction to Probability Distributions

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Probability distribution

X : Rolling a die

EVENTS	Not Rigged	Rigged
Having a 1	P(1) = 1/6	P(1) = 1/10
Having a 2	P(2) = 1/6	P(2) = 1/10
Having a 3	P(3) = 1/6	P(3) = 1/10
Having a 4	P(4) = 1/6	P(4) = 1/10
Having a 5	P(5) = 1/6	P(5) = 1/10
Having a 6	P(6) = 1/6	P(6) = 5/10

What distinguishes a rigged die from one that is not?

PROBABILITY DISTRIBUTION

The way in which probability is distributed among the different possible values of the variable.

Probability distribution

Probability function

Distribution function

Density function

Discrete variables

Continuous variables

Probability function

Function that collects the probability that the variable will take a value **equal** to a fixed x_i:

$$\boldsymbol{P(x_i) = P(X = x_i)\ \forall x_i \in E}$$

Properties:

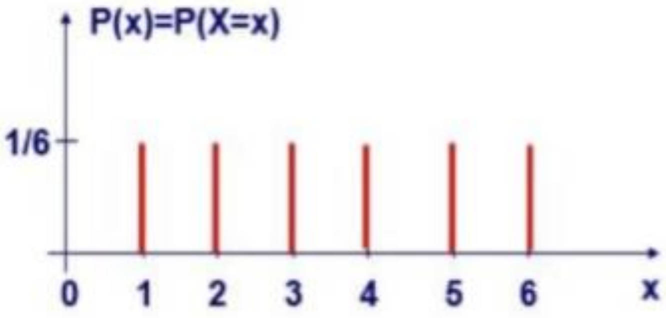
$$\boldsymbol{P(x_i) \geq 0\ \forall x_i \in E}$$

$$\boldsymbol{\sum_{\forall x_i \in E} P(x_i) = 1 \in E}$$

Probability function

X : Rolling a die

EVENTS	Fi	fi	P(X=xi)
Having a 1	149	0.149	P(X=1) = 1/6
Having a 2	178	0.178	P(X=2) = 1/6
Having a 3	177	0.177	P(X=3) = 1/6
Having a 4	184	0.184	P(X=4) = 1/6
Having a 5	152	0.152	P(X=5) = 1/6
Having a 6	160	0.160	P(X=6) = 1/6
Total	1000	1	



Distribution function

Function that collects the probability that the variable takes a **value less than or equal to** a fixed x_i:

$$\boldsymbol{F_X(x_i) = P(X \leq x_i)\ \forall x_i \in E}$$

The distribution function is a **cumulative probability** (cumulative relative frequency.

Distribution function

Properties:

$\mathbf{1})\ F(+\infty) = P(X \leq +\infty) = 1$

$\mathbf{2})\ F(-\infty) = P(X \leq -\infty) = 0$

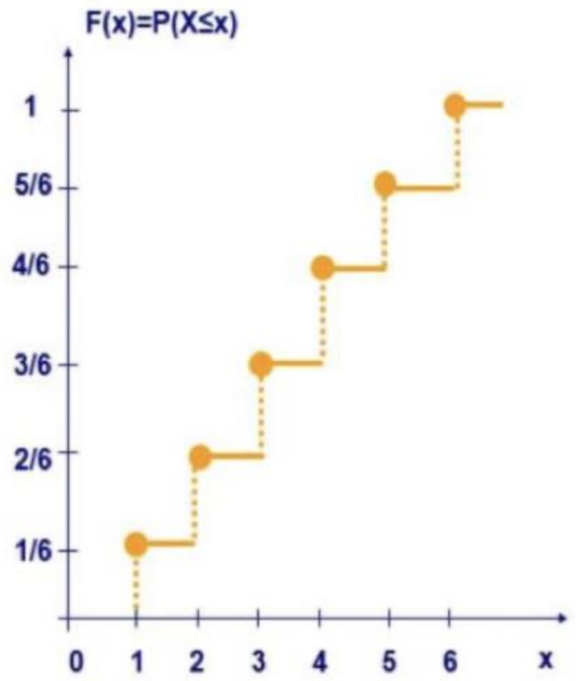
$\mathbf{3})\ F(b) - F(a) = P(a < X \leq b)$ for **discrete** variables

$\mathbf{4})\ F(b) - F(a) = P(a < X \leq b)\,) = P(a \leq X \leq b)$ for **continuous** variables

Distribution function

X : Rolling a die

EVENTS	Fi	fi	P(X=xi)	P(X≤xi)
Having a 1	149	0.149	$P(X=1) = 1/6$	$P(X\leq 1) = 1/6$
Having a 2	178	0.178	$P(X=2) = 1/6$	$P(X\leq 2) = 2/6$
Having a 3	177	0.177	$P(X=3) = 1/6$	$P(X\leq 3) = 3/6$
Having a 4	184	0.184	$P(X=4) = 1/6$	$P(X\leq 4) = 4/6$
Having a 5	152	0.152	$P(X=5) = 1/6$	$P(X\leq 5) = 5/6$
Having a 6	160	0.160	$P(X=6) = 1/6$	$P(X\leq 6) = 6/6$
Total	**1000**	**1**		



Density function

The density function is the derivative of the distribution function. Let f(x) be the derivative of F(x):

$$\boldsymbol{f_X(x) = \frac{dF_X(x)}{dx} = \lim_{\Delta x \to 0} \frac{F(x+\Delta x) - F(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x < X \leq x + \Delta x)}{\Delta x}}$$

Then:

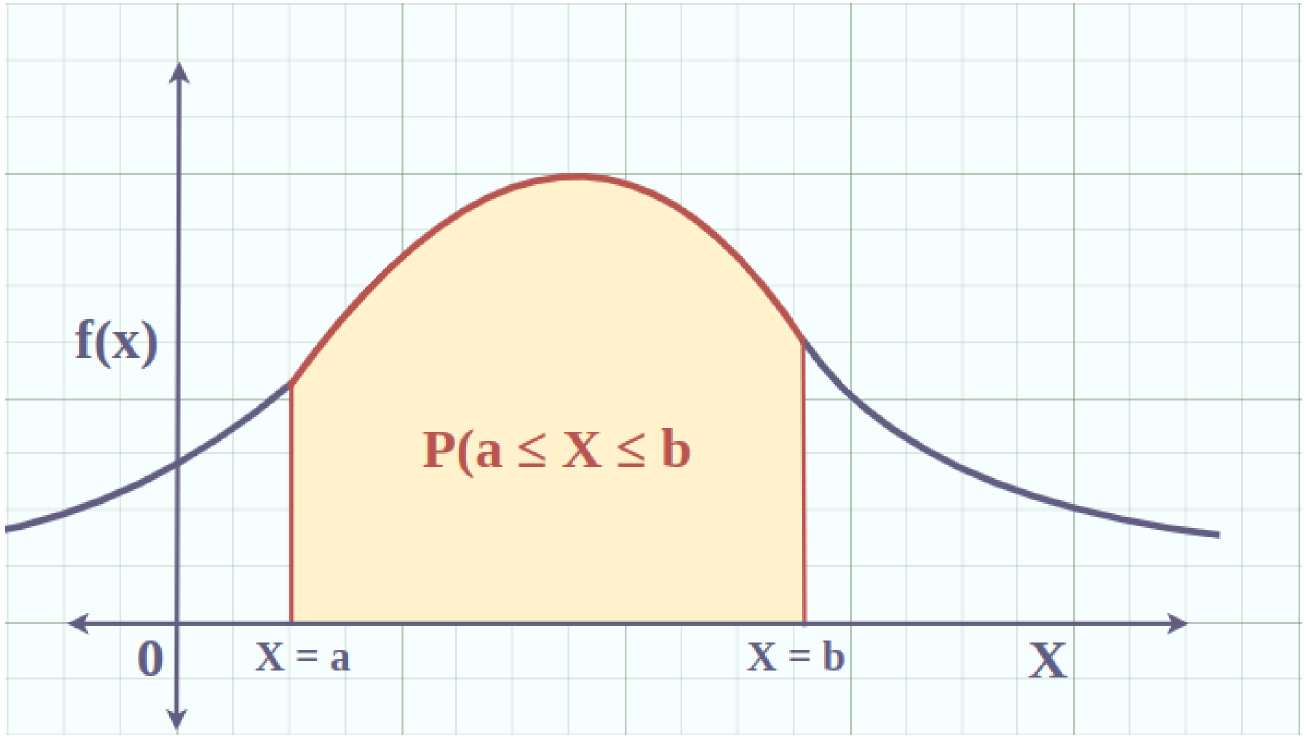
$$\boldsymbol{P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx}$$

Density function

Properties:

$$1)\ f(X) \geq 0\ \forall x \in R$$

$$2)\ \int_{-\infty}^{+\infty} f(x)dx = 1$$



Discrete VS Continuous variables

Discrete variables

- **E :** discrete/ countable infinity (N)
- **Functions:** P(x) and F(X)
- **F(X)** is a staircase function (**with jumps**) and not decreasing.
- $\exists\ x \in E: P(X = x) \neq 0$

Accumulates probability in some values of the variable

Continiuous variables

- **E :** infinite (R)
- **Functions:** f(x) and F(X)
- **F(X)** is continuous (**without jumps**) and non-decreasing
- $\forall x \in E: P(X = x) = 0$

It does not accumulate probability at any specific value of the variable.

Mathematical expectation

Mathematical expectation is used to determine the true **population** mean and variance **without sampling error**.

It can be calculated if one of the three functions (probability, distribution, and density) is known.

Population mean E(x)

X: random variable
h(X): function of X
E(h(X)): mathematical expectation of h(X). Average value of the function.

If X is **discrete**:

$$E\big(h(X)\big) = \sum_{i}^{N} h(x_i) \cdot P(X = x_i)$$

If X is **continuous**:

$$E\big(h(X)\big) = \int_{-\infty}^{+\infty} h(x) \cdot f(x)dx$$

Population mean E(x)

Properties:

$$1)\ Y = a \pm X \rightarrow m_y = a \pm m_x$$

$$2)\ Y = bX \rightarrow m_y = bm_x$$

$$3)\ Y = a \pm bX \rightarrow m_y = a \pm bm_x$$

$$4)\ Y = aX_1 \pm bX_2 \rightarrow m_y = am_{x1} \pm bm_{x2}$$

Changes in the origin and scale of measurement of the random variable result in the same changes in the value of the mean.

a and b are always constant

Population variance E(x-2)

X: random variable

h(X): $(\boldsymbol{x}-\boldsymbol{m})^{2}$

$$\boldsymbol{E}((\boldsymbol{x}-\boldsymbol{m})^{2}) = \sum (x_i - m)^2 \cdot P(X = x_i) = \sigma^2$$

Population variance E(x-2)

Properties:

$$\mathbf{1)}\ \sigma^2(aX) = a^2 \cdot \sigma^2(X)$$

$$\mathbf{2)}\ \sigma^2(a \pm X) = \sigma^2(X)$$

where X and Y are components of an independent two-dimensional random variable

$$\mathbf{3)}\ \sigma^2(aX \pm bY) = a^2\sigma^2(X) + b^2\sigma^2(X) \pm 2ab \cdot cov(X,Y)$$

where $cov(X,Y) = E[(X - m_x)\cdot(X - m_y)]$

$$\mathbf{4)}\ \sigma^2(X \pm Y) = \sigma^2(X) + \sigma^2\,(Y)$$

where $cov(X,Y) = 0$

Population CA and CC

Coefficient of Skewness (CA) Central moment of order 3

$$CA = \frac{E(X-m)^3}{\sigma^3}$$

Curtosis coefficient (CC) Central moment of order 4

$$CC = \frac{E(X-m)^4}{\sigma^4}$$

E1

An engine manufacturer produces 30% defective engines. The cost is €24 and the selling price is €54. If the engine is defective, the amount charged must be refunded and compensation of €36 must be paid.

a) Calculate the average profit per engine.
b) A control test can be performed, which costs €12 and determines with certainty whether or not the engine is defective. Consider whether it is cost-effective to perform this test.
c) An alternative test B can be used, which leads to erroneous conclusions in 10% of cases (whether the engine is correct or defective). Calculate the maximum price that can be paid for such a test.

E2

A contractor must choose between two projects. The first promises a profit of $240,000, with a probability of 0.75, or a loss of 60,000€ (due to strikes and other delays), with a probability of 0.25. the second project promises a profit of 360,000€ with a probability of 0.5 or a loss of 90,000€ with a probability of 0.5.

a) Which project should the contractor choose if he wants to maximize his expected profit?
b) Which project would he likely choose if his business was struggling and would go bankrupt unless he made a profit of 300,000 € on his next project?